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DELIVERY SYSTEMS FOR STENTS HAVING PROTRUDING FEATURES

Abstract

Delivery systems for expandable elements, such as stents or scaffolds having spikes, flails, or other protruding features for penetrating target tissue and/or delivering drugs within a human patient are described along with associated methods for using such systems. The delivery systems can be provided with a stent that is positioned over an inflatable balloon for expansion and delivery of the stent to a target delivery location. By positioning the stent over and about the inflatable balloon, the stent is ready to be expanded by the balloon immediately upon unsheathing with respect to the outer shaft. Additionally or alternatively, a stent can be positioned in an axially offset arrangement with respect to a balloon to reduce the need for space required by overlapping components.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/135,034, entitled “DELIVERY SYSTEMS FOR STENTS HAVING PROTRUDING FEATURES”, filed Apr. 14, 2023, which is a divisional of U.S. patent application Ser. No. 16/593,842, entitled “DELIVERY SYSTEMS FOR STENTS HAVING PROTRUDING FEATURES,” filed Oct. 4, 2019, which claims the benefit of U.S. Provisional Application No. 62/742,852, entitled “DELIVERY SYSTEMS FOR STENTS HAVING PROTRUDING FEATURES,” filed Oct. 8, 2018, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present description relates generally to delivery systems for expandable elements, such as stents or scaffolds having spikes, flails, or other protruding features for penetrating target tissue and/or delivering drugs within a human patient.

BACKGROUND

[0003] A variety of devices can be used to deliver drugs at desired treatment locations within a patient. For example, a stent, such as a drug-eluting stent (DES), can be positioned at the location of a stenosis (arterial narrowing) caused by arteriosclerosis. DESs generally include a drug containing polymer coated over a metal stent or scaffold, or a bioresorbable stent or scaffold composed of a drug-containing polymer. After a DES is delivered to a treatment location within a body lumen (e.g., vessel), it is expanded against a wall of the body lumen (e.g., a vessel wall) and the drug is released via direct contact with the wall. Direct delivery of the drug to the vessel wall enables significantly lower doses than those required via other delivery means (e.g., pills or injections).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 shows a partially schematic side view of an example of a delivery system.

[0005] FIG. 2 shows a cross-sectional view of an example region of the delivery system of FIG. 1 taken along line 2-2.

[0006] FIG. 3 shows an enlarged, partially schematic side view of a distal portion of the delivery system of FIG. 1 with an unconstrained stent.

[0007] FIG. 4 shows a perspective view of an example of a stent.

[0008] FIG. 5 shows a side view of the stent of FIG. 4.

[0009] FIG. 6 shows a front view of the stent of FIG. 4.

[0010] FIG. 7 shows a perspective view of an example of a connector end of the stent of FIG. 4.

[0011] FIG. 8 shows a perspective view of another example of a connector end of the stent of FIG. 4.

[0012] FIG. 9 shows a perspective view of another example of a connector end of the stent of FIG. 4.

[0013] FIG. 10 shows a perspective view of another example of a connector end of the stent of FIG. 4.

[0014] FIG. 11 shows a side view of an example of a delivery system in a first stage of deployment.

[0015] FIG. 12 shows a side view of the delivery system of FIG. 11 in a second stage of deployment.

[0016] FIG. 13 shows a side view of the delivery system of FIG. 11 in a third stage of deployment.

[0017] FIG. 14 shows a side view of the delivery system of FIG. 11 in a fourth stage of deployment.

[0018] FIG. 15 shows a cross-sectional view of an example region of a delivery system.

[0019] FIG. 16 shows a side view of another example of a delivery system in a first stage of deployment.

[0020] FIG. 17 shows a side view of the delivery system of FIG. 16 in a second stage of deployment.

[0021] FIG. 18 shows a side view of the delivery system of FIG. 16 in a third stage of deployment.

[0022] FIG. 19 shows a side view of the delivery system of FIG. 16 in a fourth stage of deployment.

[0023] FIG. 20 shows a side view of the delivery system of FIG. 16 in a fifth stage of deployment.

[0024] FIG. 21 shows a side view of another example of a delivery system in a first stage of deployment.

[0025] FIG. 22 shows a side view of the delivery system of FIG. 21 in a second stage of deployment.

[0026] FIG. 23 shows a side view of the delivery system of FIG. 21 in a third stage of deployment.

[0027] FIG. 24 shows a side view of the delivery system of FIG. 21 in a fourth stage of deployment.

[0028] FIG. 25 shows a side view of an example of a delivery system in a first stage of deployment.

[0029] FIG. 26 shows a side view of the delivery system of FIG. 25 in a second stage of deployment.

[0030] FIG. 27 shows a side view of the delivery system of FIG. 25 in a third stage of deployment.

[0031] FIG. 28 shows a side view of the delivery system of FIG. 25 in a fourth stage of deployment.

[0032] FIG. 29 shows a side view of an example of a delivery system in the delivery state within a body lumen.

[0033] FIG. 30 shows a cross-sectional view of the delivery system of FIG. 29 in a deployed state.

[0034] FIG. 31 shows a cross-sectional view of the delivery system of FIG. 29 with a balloon and a stent in an expanded state within a body lumen.

[0035] FIG. 32 shows a cross-sectional view of a region of the delivery system of FIG. 29 with a stent in a treatment state and a balloon in a collapsed state that allows for fluid flow.

[0036] In one or more implementations, not all of the depicted components in each figure may be required, and one or more implementations may include additional components not shown in a figure. Variations in the arrangement and type of the components may be made without departing from the scope of the subject disclosure. Additional components, different components, or fewer components may be utilized within the scope of the subject disclosure.

DETAILED DESCRIPTION

[0037] The detailed description set forth below is intended as a description of various implementations and is not intended to represent the only implementations in which the subject technology may be practiced. As those skilled in the art would realize, the described implementations may be modified in various different ways, all without departing from the scope of the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature and not restrictive.

[0038] The following disclosure describes various embodiments of delivery systems for expandable structures, such as stents or scaffolds, having spikes, flails, or other protruding features for penetrating target tissue and/or delivering drugs within a human patient, and associated devices and methods. The delivery systems can be configured to deliver and position expandable structures within a body lumen (e.g., vessel). In addition, these delivery systems can also be configured to deploy and expand the expandable structures in the body lumen. The delivery systems can further be configured to engage with the expanded structure and collapse the structure for removal from the body lumen. In some embodiments, the delivery systems can be configured to deliver another expandable structure or the same expandable structure to another body lumen, or the same body lumen, in a single procedure or during a plurality of procedures. Such delivery systems are expected to simplify and expedite transluminal procedures to more effectively deliver and position expandable structures within target tissues. The delivery systems can be used with more than one procedure, such as deployment of an expandable structure, when configured to re-capture the deployed expandable structure.

[0039] In particular, delivery systems described herein can be provided with a stent that is positioned over an inflatable balloon for expansion and delivery of the stent to a target delivery location. By positioning the stent over and about the inflatable balloon, the stent is ready to be expanded by the balloon immediately upon unsheathing with respect to the outer shaft. Additionally or alternatively, a stent can be positioned in an axially offset arrangement with respect to a balloon to reduce the need for space required by overlapping components.

[0040] Certain details are set forth in the following description and FIGS. **1-32** to provide a thorough understanding of various embodiments of the disclosure. To avoid unnecessarily obscuring the description of the various embodiments of the disclosure, other details describing well-known structures and systems often associated with expandable structures, protruding features, and the components or devices associated with the manufacture of such structures are not set forth below. Moreover, many of the details and features shown in the figures are merely illustrative of particular embodiments of the disclosure. Accordingly, other embodiments can have other details and features without departing from the spirit and scope of the present disclosure. A person of ordinary skill in the relevant art will therefore understand that the present technology, which includes associated devices, systems, and procedures, may include other embodiments with additional elements or steps, and/or may include other embodiments without several of the features or steps shown and described below with reference to FIGS. **1-32**. Furthermore, various embodiments of the disclosure can include structures other than those illustrated in the figures and are expressly not limited to the structures shown in the figures.

[0041] FIG. **1** shows a partially schematic side view of a delivery system **100** for a stent in a delivery state (e.g., low-profile or collapsed configuration). The delivery system **100** includes an outer shaft **120** (e.g., a catheter) having one or more lumens for containing an inner shaft **110** and/or a guidewire **162**. In some embodiments, the outer shaft **120** may also include one or more layers. In these embodiments, for example, the layers of the outer shaft **120** can include an inner layer, an outer layer, a liner, or a combination thereof. Each of the layers can be formed from materials including a polymer, high-density polyethylene (HDPE), polytetrafluoroethylene, silicone, Pebax® (polyether block amide) or a combination thereof. In some embodiments, each of the layers of the outer shaft **120** are formed from the same material. In other embodiments,

however, one or more of the layers may be formed from different materials.

[0042] The inner shaft **110** can extend from a connector **150**, through the outer shaft **120**, and beyond the distal portion **120b** of the outer shaft **120**. The inner shaft **110** can be formed as a tubular structure (with or without a slit), such as a coiled tube, a braided tube, a reinforced tube, or a combination thereof, and may be constructed of a polymer material, such as a polyimide. The delivery system **100** can include a guidewire within the inner shaft **110** and accessible at a proximal end of the delivery system **100**.

[0043] In the detailed view of the distal portion **100b** of the delivery system **100**, a tip **115** (e.g., an atraumatic tip) is disposed on a distal terminal end of the inner shaft **110**. As illustrated, the tip **115** is adjacent to a distal terminal end of the outer shaft **120**. At least a portion of the tip **115** can have the same cross-sectional dimension as the outer shaft **120**, or the tip **115** may have a different cross-sectional dimension. In some embodiments, a distal end **115b** of the tip **115** is tapered such that the distal end **115b** has a smaller cross-sectional dimension compared to a proximal end **115a** of the tip. Distal and/or proximal edges of the tip **115** may be curved/rounded so as to prevent the tip **115** from getting caught (e.g., stuck) on other portions of the delivery system **100** during delivery, positioning, deployment, etc. The tip **115** can be formed of the same material(s) as the outer shaft **120**. In other embodiments, however, the tip **115** can be formed from different material(s) than the outer shaft **120**.

[0044] The inner shaft **110** and the outer shaft **120** can be sized and shaped for intravascularly accessing a target site (e.g., treatment site) of the patient. In some embodiments, for example, the outer shaft **110** has a length of about 150 cm to about 180 cm and a suitable cross-sectional dimension for positioning within a subject's vasculature. The length of the inner shaft **110** can be a working length, such as a length that can be positioned within a subject's vasculature. In some embodiments, for example, the working length is about 70 cm to about 300 cm, about 150 cm to about 250 cm, or about 70 cm, about 80 cm, about 90 cm, about 100 cm, about 110 cm, about 120 cm, about 130 cm, about 140 cm, about 150 cm, about 160 cm, about 170 cm, about 180 cm, about 190 cm, about 200 cm, about 210 cm, about 220 cm, about 230 cm, about 240 cm, about 250 cm, about 260 cm, about 270 cm, about 280 cm, about 290 cm, or about 300 cm. In other embodiments, the outer shaft **120** has a length of about 130 centimeters (cm) to about 140 cm and a cross-sectional dimension of about 4 French, about 5 French, or about 6 French. The length of the outer shaft **120** can be a working length, such as a length that can be positioned within a subject's vasculature. In some embodiments, the working length is about 50 cm to about 200 cm, about 100 cm to about 150 cm, or about 50 cm, about 60 cm, about 70 cm, about 80 cm, about 90 cm, about 100 cm, about 110 cm, about 120 cm, about 125 cm, about 130 cm, about 135 cm, about 140 cm, about 145 cm, about 150 cm, about 155 cm, about 160 cm, about 170 cm, about 180 cm, about 190 cm, or about 200 cm.

[0045] In the detailed view of the proximal portion **100a** of the delivery system **100** in FIG. 1, a proximal end **120c** of the outer shaft is coupled to an outer shaft hub **140**. In the illustrated embodiment, the outer shaft hub **140** is coupled to the outer shaft **120** (e.g., via bonding). In other embodiments, however, the proximal end **120c** of the outer shaft is directly coupled to the outer shaft hub **140**.

[0046] The outer shaft hub **140** is further coupled to a connector **150** (e.g., y-connector) having a lumen extending therethrough (not shown). In particular, a distal end **150b** of the connector **150** can be coupled to the outer shaft hub **140** via a mating feature and a receiving feature (not shown). The mating and receiving features can be coupled to the proximal portion of the outer shaft **120** or the distal end **150b** of the connector **150**. The connector **150** further includes a port **152** extending radially and/or longitudinally therefrom. The delivery system **100** can optionally include a hemostasis connector **170** coupled to a proximal end **150a** of the y-connector **150**. While the proximal end **120c** is illustrated with particular components in a particular arrangement, it will be understood that additional or fewer components can be included in similar or other arrangements to

meet the needs of the system.

[0047] The delivery system **100** is configured to carry a stent, discussed further herein, in a delivery/collapsed state within a distal portion of the outer shaft **120**. The stent can be at least partially ensheathed by the outer shaft **120**. In some embodiments, the stent can be fixedly or removably coupled to the inner shaft **110**. Although the delivery system **100** is illustrated as a delivery system for stents, it will be appreciated that embodiments of the present technology can also include cages, meshes, balloons, membranes, tubular structures, circumferential bodies, expandable elements, expandable membranes, expandable structures, expandable tubular structures, and circumferentially expandable catheter tips with and without guidewire lumens.

[0048] FIG. 2 shows a cross-sectional view of a region of the delivery system **100** of FIG. 1 taken along line 2-2. As illustrated in FIG. 2, the inner shaft **110** can be at least partially disposed within a lumen of the outer shaft **120** and the guidewire **162** can be at least partially disposed within a lumen of the inner shaft **110**. In some embodiments, the outer shaft **120**, the inner shaft **110**, and/or the guidewire **162** each have a circular cross-sectional shape. In other embodiments, however, the outer shaft **120**, the inner shaft **110**, and/or the guidewire **162** can have other cross-sectional shapes, such as an ovoid shape, a “C” shape, a rectangular shape, a triangular shape, or the like.

[0049] The guidewire **162** and the inner shaft **110** can be positioned within the lumen of the outer shaft **120** in any configuration, such as anteriorly and posteriorly as illustrated, or medially and laterally. Furthermore, the guidewire **162** and the inner shaft **110** can be positioned in the lumen of the outer shaft **120** with respect to one another as illustrated, or the guidewire **162** can be positioned outside the inner shaft **110**. A fluid pathway can be defined within the lumen of the inner shaft **110**, for example along the length of the guidewire **162**. The fluid pathway can connect to and/or be accessible by the port **152** of the connector **150**.

[0050] FIG. 3 shows a side view of a distal portion **100b** of the delivery system **100** of FIG. 1 in a deployed state. In the illustrated embodiment, a stent **190** extends over a balloon **180** and is coupled to the delivery catheter shaft and has been unsheathed from the distal portion **120b** of the outer shaft. A proximal visualization marker **192** is disposed on the stabilizing wire **160** near a proximal portion of the stent **190** and distal visualization markers **197** are disposed on a distal end **190c** of the stent. In some embodiments, the proximal visualization marker **192** and/or the distal visualization marker **197** may be disposed on the stabilizing wire **160**. The visualization markers **192** and/or **197** can be formed from any material that can be visualized while the stent **190** is intravascularly positioned (e.g., within a target blood vessel). In one embodiment, for example, the visualization markers **192** and/or **197** are radiopaque markers. The stabilizing wire **160** can be connected to the inner shaft **110**, such that movement of the inner shaft **110** correspondingly urges the stent **190** via the stabilizing wire **160**, as discussed further herein. Alternatively, the stabilizing wire **160** can be independently movable relative to the inner shaft **110**, as discussed further herein.

[0051] The tip **115** is disposed on a terminal end **110c** of the inner shaft **110** and can surround the terminal end **110c** extending proximally along the distal portion **110b** and/or distally from the terminal end **110c**. The inner shaft **110** extends distally from the distal portion **120b** of the outer shaft **120**, through a lumen of the stent **190**, and, optionally, extends distally from the distal end of the stent **190**. In the deployed configuration, protruding features **194** extend radially from a longitudinal axis of the stent **190**, as discussed further herein.

[0052] The inner shaft **110** can also include an inflatable balloon (not shown), as discussed further herein. The inflatable balloon can be axially overlapping with the stent **190**, distal to the stent **190**, or proximal to the stent **190** while the stent **190** is in a delivery state (e.g., low-profile or collapsed configuration) within the outer shaft **120** and/or while the stent **190** is initially deployed from the delivery state.

[0053] The guidewire **162** can extend through the inner shaft **110** and beyond the tip **115**.

Accordingly, the guidewire **162** can be advanced ahead of other portions of the delivery system **100**. The inner shaft **110**, the stent **190**, and the outer shaft **120** can be advanced over the guidewire

162 until the stent **190** is aligned with a desired target delivery location. The length of the guidewire **162** that overlaps other portions of the delivery system **100** can be within the inner shaft **110**, so that it does not interfere with any other components of the delivery system **100**.

[0054] As shown in FIGS. **4-6**, the expandable stent **190** is provided with a frame **191** and multiple protruding features **194**. The frame **191** and the protruding features **194** can be configured to radially expand after the stent **190** has been unsheathed from the outer shaft **120**. The stent **190** can be self-expanding upon release from a constraint. Additionally or alternatively, the stent **190** can be expandable by radial forces applied from a balloon that is inflated while within the stent **190**. The frame **191** can include multiple struts **195** arranged in a pattern that supports compression, expansion, flexibility, and bendability of the stent **190**. The frame **191** can form a generally cylindrical shape along at least a portion of the stent **190**. At least a portion of each protruding feature **194** can extend at least partially distally from the frame **191** (e.g., towards the distal end **190c**). For example, at least a portion of each protruding feature **194** can extend parallel to a longitudinal axis of the stent **190**. At least a portion (e.g., terminal end portion) of each protruding feature **194** can extend at least partially radially away from the frame **191**. For example, at least a portion of each protruding feature **194** can extend radially outwardly (e.g., perpendicular to) the longitudinal axis of the stent **190**. With at least a portion of each protruding feature **194** extending distally from the frame **191**, the protruding features **194** can be readily retracted into the outer shaft **120** by folding down and extending distally when the outer shaft **120** is advanced from a proximal side of the stent **190** in a distal direction over the stent **190**. The protruding features **194** can optionally include drugs for delivery to a target delivery location upon expansion of the stent **190**. However, it will be understood that a stent **190** can omit drugs for delivery and treat a target delivery location by penetrating tissue with the protruding features **194**.

[0055] The frame **191**, struts **195**, and/or protruding features **194** can be composed of or formed from a variety of materials including, e.g., nitinol, cobalt chromium, stainless steel, any of a variety of other metals or metal alloys, or a combination thereof. The frame **191**, struts **195**, and/or protruding features **194** may also be composed of or formed from bioresorbable biodegradable, nanoporous or non-bioresorbable, non-biodegradable, non-nanoporous materials including, e.g., one or more polymers, nitinol, plastic materials, etc., or a combination thereof. In some embodiments, the frame **191** and the struts **195** can be formed from a bioresorbable material and the protruding features **194** can be formed from a non-bioresorbable material, such as nitinol. In these embodiments, the protruding features **194** can remain engaged with or penetrating a portion of the body lumen after the expanded frame **191** and struts **195** bio-resorb. After the expanded frame **191** and struts **195** bio-resorb, the body lumen where the stent **190** had been expanded is no longer partially occluded by the frame **191** and the struts **195** allowing for larger volumes of fluids, such as aqueous pharmaceutical compositions, to pass through the body lumen and contact the luminal wall. The protruding features **194** may also be formed of a bio-resorbable material and, once the stent **190** has bio-resorbed, the spaces in the body lumen wall vacated by the protruding features **194** can be contacted by the fluids passing through the body lumen. In this way, the stent **190** can increase a surface area of the body lumen wall contacted by the fluid.

[0056] The protruding features **194** may also be carried by more than one strut **195**, the frame **191**, or a combination thereof. The protruding features **194** may be integrally formed with the struts **195**, for example by bending or twisting a portion of one or more struts and/or the frame **191** away from a longitudinal axis of the stent **190** or, alternatively, the protruding features **194** may be separate, discrete components that are attached to desired locations along the struts **195** and/or the frame **191**.

[0057] The stent **190** can include an anchor portion **196** that securely connects to a component for controlling, positioning, and/or adjusting the stent **190**. For example, the anchor portion **196** can securely connect the stent **190** to the inner shaft **110**. Alternatively, the anchor portion **196** can securely connect the stent **190** to the stabilizing wire **160**. The anchor portion **196** can be offset

from a central axis of the stent **190**. For example, the anchor portion **196** can be radially aligned with, adjacent to, or near a portion of the frame **191** of the stent **190**. The frame **191** of the stent **190** can be connected to the anchor portion **196** by an intermediate portion **193**. The intermediate portion **193** can include multiple struts that may have varying widths to aide in column strength for deploying and retraction that extend from different portions of the frame **191**, for example connecting to different circumferential portions at an end of the frame **191**. The struts of the intermediate portion **193** can extend to the same or different axial locations along the anchor portion **196**. The arrangement of the struts of the intermediate portion **193** can maintain an open central space along the entire length of the stent **190**.

[0058] As shown in FIGS. **7-10**, the anchor portion **196** can be formed with one or more of a variety of arrangements. As shown in FIG. **7**, the anchor portion **196** can include multiple ribs **902** extending circumferentially from different axial locations along the anchor portion **196**. The ribs **902** can be positioned at different axial locations to provide multiple points of contact with a positioner, such as the inner shaft **110** and/or the stabilizing wire **160**.

[0059] As shown in FIG. **8**, the anchor portion **196** can include different portions that extend in different directions. For example, the anchor portion **196** can include longitudinal portions **904** and circumferential portions **906**. Axially adjacent pairs of the longitudinal portions **904** can be connected together by a corresponding circumferential portion **906**. Likewise, axially adjacent pairs of the circumferential portions **906** can be connected together by a corresponding longitudinal portion **904**. Different longitudinal portions **904** can have different circumferential positions to surround a coupled positioner at different circumferential positions thereon.

[0060] As shown in FIG. **9**, the anchor portion **196** can include a helical winding **908**. For example, the anchor portion **196** can wind helically about a central space configured to receive the positioner therein. The helical winding **908** can include multiple (e.g., 2, 3, 4, 5, 6, 7, 8, or more than 8) turns. The helical winding **908** can include, in cross-section, a shape that provides a flat inner side for engaging the positioner while maintaining a low profile.

[0061] As shown in FIG. **10**, the anchor portion **196** can include an arrangement of multiple struts **910**. The struts **910** can define a generally cylindrical shape for receiving and coupling to a positioner. The struts **910** can extend longitudinally and/or circumferentially about the space for receiving the positioner. The struts **910** can form any number of cells, which can vary in length and/or width relative to each other.

[0062] The anchor portion **196** can securely connect the stent **190** to a positioner, such as the inner shaft **110** and/or the stabilizing wire **160**. For example, the anchor portion **196** can be pressed onto the positioner. By further example, the anchor portion **196** can be bonded to the positioner. Additionally or alternatively, a sleeve can be provided about at least a portion of the anchor portion **196** and/or the positioner. For example, a tube, such as shrink tubing molded from one or more flexible materials, including polyurethane and Pebex® (e.g., Pebex® 35D), can be provided as a sleeve over the anchor portion **196** and/or the positioner. Additionally or alternatively, the stabilizing wire **160** can be connected to the inner shaft **110** by one or more of a variety of methods, including laser welding, bonding, crimping, swaging, reflowing, etc. Additionally or alternatively, the anchor portion **196** can removably or reversibly connect the stent **190** to a positioner. For example, the anchor portion **196** can be provided with one or more detachment mechanisms (e.g., electrolytic, mechanical, or chemical) for controllably separating the stent **190** from the positioner. As such, the stent **190** can be controllably detached and left at a target delivery location.

[0063] Methods described herein provide delivery of the stent **190** to a target delivery location by operation of the delivery system **100**. While methods in their various stages are discussed and illustrated herein, it will be understood that multiple variations of each method are also contemplated. For example, the methods can be performed in various orders of operations, with additional operations, or with fewer operations.

[0064] As shown in FIGS. **11-14**, a delivery system **100** can be provided with a stent **190** that is

positioned over an inflatable balloon **180** for expansion and delivery of the stent **190** to a target delivery location. By positioning the stent **190** over and about the inflatable balloon **180**, the stent **190** is ready to be expanded by the balloon **180** immediately upon unsheathing with respect to the outer shaft **120**.

[0065] As shown in FIG. **11**, the delivery system **100** is provided with the outer shaft **120** covering or ensheathing other components of the delivery system **100**. For example, the outer shaft **120** can extend to the tip **115** positioned at a distal end of the inner shaft **110**. The inner shaft **110** can extend within the outer shaft **120**, with a length thereof accessible proximal to a proximal end of the outer shaft **120** (e.g., at the outer shaft hub **140**). Additionally or alternatively, the connector **150** can be accessible proximal to a proximal end of the outer shaft **120** (e.g., at the outer shaft hub **140**). As discussed above, a guidewire can be advanced ahead of the tip **115** (e.g., through the inner shaft **110**) to provide a pathway for advancement of other components of the delivery system **100**.

[0066] As shown in FIGS. **12** and **13**, the outer shaft **120** can be moved to unsheath the stent **190** and other components of the delivery system **100**. For example, once the distal region of the delivery system **100** is positioned at a desired location, the outer shaft **120** is configured to be at least partially proximally retracted relative to the inner shaft **110** by retracting the outer shaft hub **140** relative to the connector **150**. Once the outer shaft **120** is partially retracted, at least a portion of the stent **190** and/or the balloon **180** is unsheathed and protruding features **194** of the stent **190** are configured to radially expand outwardly away from the inner shaft **110**.

[0067] As used herein, movement of various components can be relative to other components of the delivery system **100** and/or relative to a position apart from the delivery system **100** (e.g., a position within the anatomy of the patient, target delivery location, and/or tissue). The directions “proximal” and “distal” can be with respect to the delivery system **100**, a component thereof, and/or a position apart from the delivery system **100**. For example, movement of the guidewire **162** can be relative to the outer shaft **120**, the inner shaft **110**, the stent **190**, and/or the balloon **180**. It will be understood that while the guidewire **162** moves, the outer shaft **120**, the inner shaft **110**, the stent **190**, and/or the balloon **180** can be stationary, moving in the same direction (e.g., at a different speed), or moving in a different (e.g., opposite) direction. It will be further understood that while the outer shaft **120**, the inner shaft **110**, the stent **190**, and/or the balloon **180** moves, the guidewire **162** can be stationary, moving in the same direction (e.g., at a different speed), or moving in a different (e.g., opposite) direction. By further example, movement of the outer shaft **120** can be relative to the inner shaft **110**, the stent **190**, and/or the balloon **180**. It will be understood that while the outer shaft **120** moves, the inner shaft **110**, the stent **190**, and/or the balloon **180** can be stationary, moving in the same direction (e.g., at a different speed), or moving in a different (e.g., opposite) direction. By further example, movement of the inner shaft **110**, the stent **190**, and/or the balloon **180** can be relative to outer shaft **120**. It will be understood that while the inner shaft **110**, the stent **190**, and/or the balloon **180** move, the outer shaft **120** can be stationary, moving in the same direction (e.g., at a different speed), or moving in a different (e.g., opposite) direction.

[0068] As shown in FIG. **14**, the outer shaft **120** has been retracted and the stent **190** is unsheathed. The stabilizing wire **160**, connected to the inner shaft **110** by the anchor portion **196**, is configured to engage with the proximal end of the stent **190** and control the position of the stent **190** during and after retraction of the outer shaft **120**. Accordingly, the position of the stent **190** is maintained with respect to the inner shaft **110**, including the balloon **180**. For example, while some adjustment of the length and/or axial position of the stent **190** may occur during radial expansion of the stent **190**, it will be understood that the stabilizing wire **160** can maintain the position of at least a portion of the stent **190** to be around and axially aligned with at least a portion of the balloon **180**. The balloon **180** can have an axial length that is greater than the axial length of the stent **190**, so that an entirety of the stent **190** is overlapping with the balloon **180**. As shown in FIG. **14**, the stabilizing wire **160** can connect the stent **190** to a portion of the inner shaft **110** that is proximal to the balloon **180**. Additionally or alternatively, the stabilizing wire **160** can connect the stent **190** to

a portion of the inner shaft **110** that is distal to the balloon **180**.

[0069] When both the stent **190** and the balloon **180** are unsheathed by the outer shaft **120** and exposed, the balloon **180** can be inflated to expand or further expand the stent **190**. For example, an interior region of the balloon can be fluidly connected, via the inner shaft **110**, to the port **152** of the connector **150**. By providing a fluid through the port **152**, the balloon **180** can be expanded, thereby expanding or further expanding the stent **190**. The expansion with respect to target anatomy will be further discussed herein.

[0070] Following one or more of the above-described operations, the balloon **180** can be deflated. The stent **190** can be maintained for any duration of time in an expanded state. For example, the stent **190** can be maintained for a duration of time effective to provide therapeutic treatment (e.g., remodeling and/or drug delivery) to target anatomy and allows fluid flow through the expanded stent and deflated balloon where there is no fluid blockage through the treated site.

[0071] Additionally or alternatively, the delivery system **100** can be deployed at multiple locations. The stent **190** can be collapsed by moving the outer shaft **120** over the stent **190**. The stent **190** and the balloon **180** can be moved to another target location, and one or more of the above-described operations can be repeated.

[0072] Additionally or alternatively, the delivery system **100** can be removed. The stent **190** can be collapsed by moving the outer shaft **120** over the stent **190**. Components of the delivery system **100** can be removed from the patient by retracting proximally over the guidewire.

[0073] Additionally or alternatively, the stent **190** can be detached from the inner shaft **110** and left as an implant within the patient. Following detachment, other components of the delivery system **100** can be removed from the patient by retracting proximally over the guidewire.

[0074] While the delivery system **100** is shown with a stent **190** positioned over a balloon **180** in a delivery state, it will be understood that other arrangements are contemplated. For example, a stent can be positioned in an axially offset arrangement with respect to a balloon to reduce the need for space required by overlapping components. Reference is made to a delivery system **200**, as shown in FIGS. **16-20**, a delivery system **300**, as shown in FIGS. **21-24**, and a delivery system **500**, as shown in FIGS. **25-28**. While each of the delivery system **200** and the delivery system **300** is in some aspects different than the delivery system **100**, it will be understood that components and features of the delivery system **100** as described herein can apply to either or both of the delivery system **200** and the delivery system **300**. Similar or like items can perform the same function as those shown in the delivery system **100**, and the features of such items are not all discussed hereafter, for brevity.

[0075] Referring now to FIG. **15**, with further reference to FIG. **1**, a cross-sectional view of a region of a delivery system **200** is shown, wherein the delivery system **200** is similar in at least some aspects to the delivery system **100** shown in FIG. **1**. For example, the sectional view of FIG. **15** can be taken along a line that positioned similarly as the line 2-2 in FIG. **1**. As illustrated in FIG. **15**, an inner shaft **210** can be at least partially disposed within a lumen of an outer shaft **220**.

Additionally, a stiffening wire **264** is provided between the outer shaft **220** and the inner shaft **210**. The stiffening wire **264** can be of stainless steel or another material and can influence the luminal space and shaft stiffness and/or flexibility without modifying the material thickness of the shaft. A guidewire **262** can be at least partially disposed within a lumen of the inner shaft **210**. A lumen defined between the outer shaft **220** and the inner shaft **210** or within the inner shaft **210** can provide fluid communication to a balloon for inflation and deflation of the balloon. In some embodiments, the outer shaft **220**, the inner shaft **210**, the guidewire **262**, and/or stiffening wire **264** each have a circular cross-sectional shape and a single lumen. In other embodiments, however, the outer shaft **220**, the inner shaft **210**, the guidewire **262**, and/or stiffening wire **264** can have other cross-sectional shapes, such as an ovoid shape, a “C” shape, a rectangular shape, a triangular shape, or the like, with multiple lumens. For example, the stiffening wire **264** can have a shape that fits within the space between the outer shaft **220** and the inner shaft **210**. For example, the cross-

sectional shape of the stiffening wire **264** can be polygonal (e.g., rectangular) or crescent-shaped. The inner surface of the outer shaft **220** and/or the outer surface of the inner shaft **210** can have cross-sectional shapes that accommodate and/or guide the stiffening wire **264**. A support shaft **230** can also be placed circumferentially around the inner shaft **210** or outer shaft **220** to increase column strength and stiffness of the catheter region. In some embodiments, the support shaft **230** can have a larger inner diameter and be attached to the outer shaft **220** to extend the overall catheter length and accommodate a larger proximal segment of the inner catheter **210** or stiffening wire **264**. The support shaft placement can vary from the entire length of the inner shaft **210** to specific 10 cm, 20 cm, 30 cm, 40 cm segments of the inner shaft with varying gaps of 10 cm, 20 cm, 30 cm, 40 cm length to increase overall catheter stiffness. Attachment mechanisms may include bonding, reflowing, braiding, coiling, laser welding, etc. The support shaft **230** can also be made of high durometer plastics such as nylon, Pebax®, stainless steel, Nitinol, polyether ether ketone (PEEK), etc.

[0076] A stabilizing wire **260** can be coupled to a stent. The stabilizing wire **260** is slideably disposed within the outer shaft **220** and is sized and shaped to extend distally from the proximal end of the outer shaft and to extend proximally from a proximal end of a port. The stabilizing wire **260** can be formed of plastic, such as high durometer plastic including nylon, polyether ether ketone (PEEK), a metal, a metal alloy, such as nitinol, and/or combinations thereof. The stabilizing wire **260** can be configured to position the stent (not shown) at the desired treatment location and to at least generally maintain the position of the stent while the outer shaft **220** is withdrawn as described in greater detail below.

[0077] The stabilizing wire **260** can be sized and shaped to extend proximally from the proximal end of the port when the stent is positioned at the target site. For example, the stabilizing wire **260** can have a length of about 150 cm to about 180 cm and a suitable cross-sectional dimension for positioning within the patient's body lumen. The stabilizing wire **260** can have a working length (i.e., a length that can be positioned within the target body lumen) of about 70 cm to about 300 cm, about 150 cm to about 250 cm, or about 70 cm, about 80 cm, about 90 cm, about 100 cm, about 110 cm, about 120 cm, about 130 cm, about 140 cm, about 150 cm, about 160 cm, about 170 cm, about 180 cm, about 190 cm, about 200 cm, about 210 cm, about 220 cm, about 230 cm, about 240 cm, about 250 cm, about 260 cm, about 270 cm, about 280 cm, about 290 cm, or about 300 cm.

[0078] As shown in FIGS. **16-20**, the delivery system **200** can be provided with a stent **290** that is positioned proximal to an inflatable balloon **280** for expansion and delivery of the stent **290** to a target delivery location. By positioning the stent **290** proximal to the inflatable balloon **280**, the stent **290** and the balloon **280** are not overlapping (e.g., are axially offset) while in a delivery state within the outer shaft **220**, and thereby reduce the space requirements within the outer shaft **220**.

[0079] As shown in FIG. **16**, the delivery system **200** is provided with the outer shaft **220** covering or ensheathing other components of the delivery system **200**. For example, the outer shaft **220** can extend to a tip **215** positioned at a distal end of the inner shaft **210**. The inner shaft **210** can extend within the outer shaft **220**, with a length thereof accessible proximal to a proximal end of the outer shaft **220** (e.g., at the outer shaft hub **240**). Additionally or alternatively, the connector **250** can be accessible proximal to a proximal end of the outer shaft **220** (e.g., at the outer shaft hub **240**). A stabilizing wire **260** is also accessible proximal to a proximal end of the outer shaft **220** (e.g., at the outer shaft hub **240**). A guidewire can be advanced ahead of the tip **215** (e.g., through the inner shaft **210**) to provide a pathway for advancement of other components of the delivery system **200**.

[0080] As shown in FIG. **17**, the outer shaft **220** can be moved to unsheath an inflatable balloon **280**. For example, once the distal region of the delivery system **200** is positioned at a desired location, the outer shaft **220** is configured to be at least partially proximally retracted relative to the inner shaft **210** by retracting the outer shaft hub **240** relative to the connector **250**. Once the outer shaft **220** is partially retracted, at least a portion of the balloon **280** is unsheathed.

[0081] As shown in FIG. **18**, the outer shaft **220** can be further moved to unsheath a stent **290**.

Once the outer shaft **220** is further retracted, a portion of the stent **290** is unsheathed and protruding features **294** of the stent **290** are configured to radially expand outwardly away from the inner shaft **210**. As shown, the balloon **280** is positioned at a distal portion **210b** of the inner shaft **210**, and the stent **290** is positioned at a proximal portion **210a** of the inner shaft **210**. The proximal portion **210a** of the inner shaft **210** can have an outer cross-sectional dimension that is smaller than an outer cross-sectional dimension of the balloon **280**, thereby permitting the stent **290** to be collapsed onto the proximal portion **210a** in a smaller profile than would be achieved if the stent **290** were collapsed onto the balloon **280**.

[0082] As shown in FIG. **19**, the outer shaft **220** has been retracted and the stent **290** is unsheathed. The stabilizing wire **260**, accessible by a user, is configured engage with the proximal end of the stent **290** and control the position of the stent **290** during and after retraction of the outer shaft **220**. The user can secure the stabilizing wire **260** relative to the inner shaft **210** while the outer shaft **220** is retracted so that the position of the stent **290** can be maintained with respect to the inner shaft **210**, including the balloon **280**, during retraction of the outer shaft **220**.

[0083] As shown in FIG. **20**, when both the stent **290** and the balloon **280** are unsheathed by the outer shaft **220** and exposed, the stent **290** can be axially aligned with the balloon **280**. Because the inner shaft **210** extends through the stent **290**, proximal retraction of the inner shaft **210** relative to the stent **290** can achieve axially alignment of the balloon **280** with the stent **290**. The balloon **280** can have an axial length that is greater than the axial length of the stent **290**, so that an entirety of the stent **290** is overlapping with the balloon **280** when axially aligned. Additionally or alternatively, the inner shaft **210** and the outer shaft **220** can be retracted together with respect to the stent **290**.

[0084] The balloon **280** can be inflated to expand or further expand the stent **290**. For example, an interior region of the balloon can be fluidly connected, via the inner shaft **210**, to the port **252** of the connector **250**. By providing a fluid through the port **252**, the balloon **280** can be expanded, thereby expanding or further expanding the stent **290**. The expansion with respect to target anatomy will be further discussed herein.

[0085] Following one or more of the above-described operations, the balloon **280** can be deflated. The stent **290** can be maintained for any duration of time in an expanded state. For example, the stent **290** can be maintained for a duration of time effective to provide therapeutic treatment (e.g., remodeling and/or drug delivery) to target anatomy.

[0086] Additionally or alternatively, the delivery system **200** can be deployed at multiple locations. The stent **290** can be collapsed by moving the outer shaft **220** over the stent **290**. Optionally, the stent **290** can be axially realigned with the proximal portion **210a** of the inner shaft **210** prior to collapse by the outer shaft **220**. The stent **290** and the balloon **280** can be moved to another target location, and one or more of the above-described operations can be repeated.

[0087] Additionally or alternatively, the delivery system **200** can be removed. The stent **290** can be collapsed by moving the outer shaft **220** over the stent **290**. Optionally, the stent **290** can be axially realigned with the proximal portion **210a** of the inner shaft **210** prior to collapse by the outer shaft **220**. Components of the delivery system **200** can be removed from the patient by retracting proximally over the guidewire.

[0088] Additionally or alternatively, the stent **290** can be detached from the stabilizing wire **260** and left as an implant within the patient. Following detachment, other components of the delivery system **200** can be removed from the patient by retracting proximally over the guidewire.

[0089] As shown in FIGS. **21-24**, the delivery system **300** can be provided with a stent **390** that is positioned distal to an inflatable balloon **380** for expansion and delivery of the stent **390** to a target delivery location. By positioning the stent **390** distal to the inflatable balloon **380**, the stent **390** and the balloon **380** are not overlapping (e.g., are axially offset) while in a delivery state within the outer shaft **320**, and thereby reduce the space requirements within the outer shaft **320**.

[0090] As shown in FIG. **21**, the delivery system **300** is provided with the outer shaft **320** covering

or ensheathing other components of the delivery system **300**. For example, the outer shaft **320** can extend to a tip **315** positioned at a distal end of the inner shaft **310**. The inner shaft **310** can extend within the outer shaft **320**, with a length thereof accessible proximal to a proximal end of the outer shaft **320** (e.g., at the outer shaft hub **340**). Additionally or alternatively, the connector **350** can be accessible proximal to a proximal end of the outer shaft **320** (e.g., at the outer shaft hub **340**). A stabilizing wire **360** is also accessible proximal to a proximal end of the outer shaft **320** (e.g., at the outer shaft hub **340**). A guidewire can be advanced ahead of the tip **315** (e.g., through the inner shaft **310**) to provide a pathway for advancement of other components of the delivery system **300**. [0091] As shown in FIG. 22, the outer shaft **320** can be moved to unsheath an inflatable balloon **380**. For example, once the distal region of the delivery system **300** is positioned at a desired location, the outer shaft **320** is configured to be at least partially proximally retracted relative to the inner shaft **310** by retracting the outer shaft hub **340** relative to the connector **350**. Once the outer shaft **320** is partially retracted, a portion of the stent **390** is unsheathed and protruding features **394** of the stent **390** are configured to radially expand outwardly away from the inner shaft **310**. The stabilizing wire **360**, accessible by a user, is configured engage with the proximal end of the stent **390** and control the position of the stent **390** during and after retraction of the outer shaft **320**. The user can secure the stabilizing wire **360** relative to the inner shaft **310** while the outer shaft **320** is retracted so that the position of the stent **390** can be maintained with respect to the inner shaft **310**, including the balloon **380**, during retraction of the outer shaft **320**.

[0092] As shown in FIG. 23, the outer shaft **320** can be further moved to unsheath a balloon **280**. Once the outer shaft **320** is further retracted, at least a portion of the balloon **380** is unsheathed. As shown, the balloon **380** is positioned at a proximal portion **310a** of the inner shaft **310**, and the stent **390** is positioned at a distal portion **310b** of the inner shaft **310**. The distal portion **310b** of the inner shaft **310** can have an outer cross-sectional dimension that is smaller than an outer cross-sectional dimension of the balloon **380**, thereby permitting the stent **390** to be collapsed onto the distal portion **310b** in a smaller profile than would be achieved if the stent **390** were collapsed onto the balloon **380**.

[0093] As shown in FIG. 24, when both the stent **390** and the balloon **380** are unsheathed by the outer shaft **320** and exposed, the stent **390** can be axially aligned with the balloon **380**. Because the inner shaft **310** extends through the stent **390**, distal movement of the inner shaft **310** relative to the stent **390** can achieve axially alignment of the balloon **380** with the stent **390**. The balloon **380** can have an axial length that is greater than the axial length of the stent **390**, so that an entirety of the stent **390** is overlapping with the balloon **380** when axially aligned.

[0094] The balloon **380** can be inflated to expand or further expand the stent **390**. For example, an interior region of the balloon can be fluidly connected, via the inner shaft **310**, to the port **352** of the connector **350**. By providing a fluid through the port **352**, the balloon **380** can be expanded, thereby expanding or further expanding the stent **390**. The expansion with respect to target anatomy will be further discussed herein.

[0095] Following one or more of the above-described operations, the balloon **380** can be deflated. The stent **390** can be maintained for any duration of time in an expanded state. For example, the stent **390** can be maintained for a duration of time effective to provide therapeutic treatment (e.g., remodeling and/or drug delivery) to target anatomy.

[0096] Additionally or alternatively, the delivery system **300** can be deployed at multiple locations. The stent **390** can be collapsed by moving the outer shaft **320** over the stent **390**. Optionally, the stent **390** can be axially realigned with the proximal portion **310a** of the inner shaft **310** prior to collapse by the outer shaft **320**. The stent **390** and the balloon **380** can be moved to another target location, and one or more of the above-described operations can be repeated.

[0097] Additionally or alternatively, the delivery system **300** can be removed. The stent **390** can be collapsed by moving the outer shaft **320** over the stent **390**. Optionally, the stent **390** can be axially realigned with the proximal portion **310a** of the inner shaft **310** prior to collapse by the outer shaft

320. Components of the delivery system **300** can be removed from the patient by retracting proximally over the guidewire.

[0098] Additionally or alternatively, the stent **390** can be detached from the stabilizing wire **360** and left as an implant within the patient. Following detachment, other components of the delivery system **300** can be removed from the patient by retracting proximally over the guidewire.

[0099] As shown in FIGS. **25-28**, a delivery system **500** can be provided with a stent **590** that is positioned over an inflatable balloon **580** for expansion and delivery of the stent **590** to a target delivery location. By positioning the stent **590** over and about the inflatable balloon **580**, the stent **590** is ready to be expanded by the balloon **580** immediately upon partial or complete unsheathing with respect to the outer shaft **520**. Any released portion of a stent **590** can be expanded by an underlying portion of a balloon **580** that is similarly released.

[0100] As shown in FIG. **25**, the delivery system **500** is provided with the outer shaft **520** covering or ensheathing other components of the delivery system **500**. For example, the outer shaft **520** can extend to the tip **515** positioned at a distal end of the inner shaft **510**. The inner shaft **510** can extend within the outer shaft **520**, with a length thereof accessible proximal to a proximal end of the outer shaft **520** (e.g., at the outer shaft hub **540**). Additionally or alternatively, the connector **550** can be accessible proximal to a proximal end of the outer shaft **520** (e.g., at the outer shaft hub **540**). As discussed above, a guidewire can be advanced ahead of the tip **515** (e.g., through the inner shaft **510**) to provide a pathway for advancement of other components of the delivery system **500**.

[0101] As shown in FIG. **26**, the outer shaft **520** can be moved to unsheath a portion of the stent **590** and other components of the delivery system **500**. For example, once the distal region of the delivery system **500** is positioned at a desired location, the outer shaft **520** is configured to be at least partially proximally retracted relative to the inner shaft **510** by retracting the outer shaft hub **540** relative to the connector **550**. Once the outer shaft **520** is partially retracted, a portion of the stent **590** and/or the balloon **580** is unsheathed and protruding features **594** of the exposed portion of the stent **590** are configured to radially expand outwardly away from the inner shaft **510**.

[0102] The stent **590** can be fixedly attached to another component of the delivery system **500**, such as the inner shaft **510** (e.g., via an anchor portion). Alternatively, the stent **590** can be adjustably positioned relative to one or more other components of the delivery system **500**. For example, the stent **590** can be coupled to a stabilizing wire that is accessible to a user at a proximal end of the delivery system **500**, and the user can adjust a position of the stent **590** by operation of the stabilizing wire.

[0103] As shown in FIG. **27**, the outer shaft **520** has been partially retracted and the stent **590** is partially unsheathed. The stent **590** can be connected to the inner shaft **510**, for example with a stabilizing wire (not shown), as described herein. Accordingly, the position of the stent **590** can be maintained with respect to the inner shaft **510**, including the balloon **580**. For example, while some adjustment of the length and/or axial position of the stent **590** may occur during radial expansion of the stent **590**, it will be understood that the stabilizing wire can maintain the position of at least a portion of the stent **590** to be around and axially aligned with at least a portion of the balloon **580**. Additionally or alternatively, the stent **590** can be secured relative to the outer shaft **520** by locking the outer shaft hub **540** relative to the inner shaft **510** at a proximal portion of the delivery system **500**. For example, a locking member can be controllably engaged and disengaged to selectively lock relative axial positions and/or movement of the outer shaft hub **540** and the inner shaft **510**. Such a locking member, when engaged can prevent proximal retraction of the outer shaft **520** when the balloon **580** is expanded, including portions of the balloon **580** that are within the outer shaft **520**. The balloon **580** can have an axial length that is greater than the axial length of the stent **590**, so that an entirety of the stent **590** is overlapping with the balloon **580**.

[0104] An extent to which the stent **590** and/or the balloon **580** are unsheathed (e.g., partially unsheathed) can be determined by one or more of a variety of mechanisms. For example, the stent **590**, the balloon **580**, the outer shaft **520**, and/or one or more other components coupled to one or

more of the above can include a visualization marker, such as a radiopaque marker. The position of such components relative to each other and/or a target location can be determined visually, for example by an imaging technique (e.g., angiography). Additionally or alternatively, the relative positions of the stent **590**, the balloon **580**, and/or the outer shaft **520** can be determined and/or inferred by corresponding components at a proximal end of the delivery system **500**. For example, the positions the outer shaft hub **540**, the inner shaft **510**, and/or a stabilizing wire (not shown) can be compared to determine the relative positions of the outer shaft **520**, the balloon **580**, and/or the stent **590**, respectively. Appropriate markers, detents, or other indicators can be provided on the outer shaft hub **540**, the inner shaft **510**, and/or a stabilizing wire (not shown) at the proximal end of the delivery system **500** for reference by a user. For example, such markers, detents, or other indicators can be incrementally spaced apart from each other to indicate to the user a position of the outer shaft hub **540** with respect to the inner shaft **510**. Such an indication can be correlated with an extent to which the stent **590** is unsheathed.

[0105] When both the stent **590** and the balloon **580** are partially unsheathed by the outer shaft **520**, the initially exposed portion of the balloon **580** can be inflated to expand or further expand the stent **590**. For example, an interior region of the initially exposed portion of the balloon **580** can be fluidly connected, via the inner shaft **510**, to the port **552** of the connector **550**. By providing a fluid through the port **552**, the initially exposed portion of the balloon **580** can be expanded, thereby expanding or further expanding the initially exposed portion of the stent **590**. Other portions of the stent **590** and/or the balloon **580** can remain within the outer shaft **520**. Expansion can be performed, for example, while the outer shaft **520** is locked relative to the inner shaft **510** (e.g., with the outer shaft hub **540**). Such locking can prevent the outer shaft **520** from further retracting in response to forces due to expansion of the partially exposed stent **590** and/or balloon **580**. The expansion with respect to target anatomy will be further discussed herein.

[0106] Following an initial deployment, additional operations can be performed to expand the stent **590** in a subsequent stage of the same procedure. For example, a different length and/or portion of the stent **590** can be utilized in a subsequent operation. As shown in FIG. **28**, the outer shaft **520** has been further retracted and the stent **590** is further unsheathed. When both the stent **590** and the balloon **580** are more fully unsheathed by the outer shaft **520**, the more fully exposed portion of the balloon **580** can be inflated to expand or further expand the more fully exposed portion of the stent **590**, for example via the port **552**, as discussed herein. Other portions of the stent **590** and/or the balloon **580** can remain within the outer shaft **520**. As described above, expansion can be performed while the outer shaft **520** is locked relative to the inner shaft **510** (e.g., with the outer shaft hub **540**) to stabilize the system during inflation of the balloon **580**.

[0107] An extent to which the stent **590** and/or the balloon **580** are unsheathed (e.g., further unsheathed) can again be determined by one or more of a variety of mechanisms, such as those described above with respect to determining an extent of partial unsheathing.

[0108] In some embodiments, the operating length of the stent **590** that is unsheathed, exposed, and/or expanded in the operations described above can be of different lengths. For example, the operating length can be shorter in an initial stage and longer in a subsequent stage. Alternatively, the operating length can be longer in an initial stage and shorter in a subsequent stage.

[0109] Where different operating lengths are desired, the balloon **580** can optionally include multiple segments that are independently inflatable. For example, the balloon **580** can include multiple segments that are aligned at different axial locations along the inner shaft **510**. The inner shaft **510** can provide multiple lumens each connecting to corresponding ports. A fluid can be provided through a selected number of the ports to inflate only the corresponding balloon segments. For example, only the balloon segments that are outside of the outer shaft **520** can be inflated to expand corresponding portions of the stent **590**. Additionally or alternatively, the balloon segments can be in fluid communication with each other such that they are inflated in a sequence.

[0110] Between an initial expansion (e.g., the expansion illustrated in FIG. **27**) and a subsequent

expansion (e.g., the expansion illustrated in FIG. 28), the stent **590** and/or the balloon **580** can be deflated, compressed, and/or at least partially retracted into the outer shaft **520**. Alternatively, the stent **590** and/or the balloon **580** can be further exposed by unlocking the outer shaft **520** from the inner shaft **510** and allowing the outer shaft **520** to further retract in response to forces from the inflated balloon **580**.

[0111] The transition from an initial expansion to a subsequent expansion can be performed to adjust an operating length of the stent **590** to more fully address a target region. For example, an initial operating length of the stent **590** can be exposed and expanded. The user can then evaluate the effectiveness of the operation (e.g., via imaging technique such as angiography). Where the initial operating length of the stent **590** is determined to be insufficient, the stent **590** and/or the balloon **580** can be further exposed to increase the operating length of the stent **590**. Such adjustments can be made as needed until an adequate operating length is provided. It will be recognized that the ability to perform such adjustments may avoid the need to remove a stent that is discovered to be inadequate and replace it with a different stent or other device that provides an adequate operating length. By eliminating these steps, total operation time can be reduced. Additionally, a user can desire to deploy a device with an operating length that is adequately long (e.g., to span a target region) without being longer than is required (e.g., to avoid operating on regions outside the target region). It will be recognized that the user can provide a single stent **590** with an adjustable operating length to adequately address a target region that has an initially uncertain length or where the required operating length of the stent is otherwise unknown or uncertain. Such capabilities reduce the burden on the user to accurately select the devices with the correct operating length at the beginning of an operation. Furthermore, the capabilities described herein also reduce the need to provide a wide array of devices that provide different performance characteristics, as a single device or reduced number of devices can be operated as described herein to provide a desirably wide range of performance characteristics.

[0112] The transition from an initial expansion to a subsequent expansion can be performed to address different operating length requirements of different target regions. Between an initial expansion and a subsequent expansion, the stent **590** and/or the balloon **580** can be repositioned to a different location. For example, the stent **590** can be repositioned to align with a different target region. Where the new target region has a different length or other feature relative to an initial target region, the operating length of the stent **590** can be selected and/or modified accordingly to adequately address each of the target regions. It will be recognized that the ability to perform such adjustments may avoid the need to remove a stent suitable for an initial target region and replace it with a different stent that is suitable for a different target region. By eliminating these steps, total procedure time can be reduced, thereby reducing risks associated with long procedure times. Additionally, it will be recognized that the user can provide a single stent **590** with an adjustable operating length to adequately address each of different target regions despite each target region having potentially different requirements for an operating length of the stent **590**. This allows a user with greater flexibility and options throughout a procedure with a single device.

[0113] Following one or more of the above-described operations, the balloon **580** can be deflated. The stent **590** can be maintained for any duration of time in an expanded state. For example, the stent **590** can be maintained for a duration of time effective to provide therapeutic treatment (e.g., remodeling and/or drug delivery) to target anatomy and allows fluid flow through the expanded stent and deflated balloon where there is no fluid blockage through the treated site.

[0114] Additionally or alternatively, the delivery system **500** can be deployed at multiple locations. The stent **590** can be collapsed by moving the outer shaft **520** over the stent **590**. The stent **590** and the balloon **580** can be moved to another target location, and one or more of the above-described operations can be repeated.

[0115] Additionally or alternatively, the delivery system **500** can be removed. The stent **590** can be collapsed by moving the outer shaft **520** over the stent **590**. Components of the delivery system **500**

can be removed from the patient by retracting proximally over the guidewire.

[0116] Additionally or alternatively, the stent **590** can be detached from the inner shaft **510** and left as an implant within the patient. Following detachment, other components of the delivery system **500** can be removed from the patient by retracting proximally over the guidewire.

[0117] Referring now to FIGS. **29-32**, an example of a delivery system **400** is shown in various configurations to deliver, position, deploy, and/or recapture a stent. The operations described with respect to the delivery system **400** can be applied to the delivery system **100**, the delivery system **200**, the delivery system **300**, and/or the delivery system **500**. As shown in FIG. **29**, the delivery system **400** is in a delivery state within a body lumen **710** (e.g., a blood vessel) of a human patient. In this embodiment, the delivery system **400** is configured for intraluminal (e.g., intravascular) delivery through the blood vessel, (e.g., femoral artery) of a human patient. The femoral artery can be accessed by introducing a sheath (e.g., **5F** or **6F**) into the lumen of the femoral artery. The delivery system **400** is delivered into the body lumen by tracking the distal portion **410b** of the inner sheath over the guidewire and distally advancing the delivery system **400** to a desired location **720** within the vessel. In some embodiments, an angioplasty procedure is performed at the desired location **720** before the delivery system **400** is advanced to the desired location **720**.

[0118] Once the delivery system **400** is positioned at the desired location **720**, the distal portion **420b** of the outer sheath is proximally retracted to unsheath the stent **490**. In the illustrated embodiment, the body of the stent **490** is at least partially expanded when unsheathed and the protruding features **494** are collapsed. However, the protruding features **494** can be configured to expand once the distal portion **420b** of the outer sheath is retracted. In other embodiments, the stabilizing wire (not shown) can be distally advanced and fixed, such as held or pinned, or fixed at the desired location to position the stent before, during, and/or after the outer sheath is proximally retracted to deploy the stent. As illustrated, the tip **415** of delivery system **400** is positioned distally from the distal end of the stent and the inner shaft **410** remains positioned within at least a portion of the lumen of the stent **190**.

[0119] In the deployed state, the protruding features **494** of the stent **490** are configured to expand radially and are further configured to pierce the lumen wall at the desired location once the deployed stent **490** is expanded into contact with the vessel wall (see FIGS. **29** and **30**). As will be explained in greater detail below, stents and other expandable structures can be configured to at least partially self-expand, such as expanding outwardly from the collapsed/delivery state to the deployed and/or expanded state when the stents and other expandable structures are at least partially unsheathed from the outer shaft. In some embodiments, stents and other expandable structures are configured to expand when operably coupled with an expandable element or mechanism, such as a balloon. In additional embodiments, self-expanding stents and other structures are configured to further expand when coupled to the expansion mechanism. Regardless of whether the stents and other expandable structures are self-expanding or expand when coupled to the expandable element, the stents and others expandable structures can be configured to expand radially (symmetrically or asymmetrically). In some embodiments, the at least partially expanded stents and other expandable structures can be configured to position at least some of the protruding features perpendicular to the vessel wall.

[0120] As shown in FIGS. **29** and **30**, the delivery system **400** is configured for insertion of a balloon **480** coupled to the inner shaft **410**. The balloon **480** is further configured to be positioned within the stent lumen and expanded therein to further expand the stent **490** between the delivery state and the expanded, deployed state. In some embodiments, the balloon **480** can be coated with a drug-delivery coating and a drug, such as the coatings and drugs described herein. As further discussed elsewhere herein, the stent **490** can be operatively coupled to an actuation mechanism, such as a mechanical actuation mechanism (e.g., stabilizing wire, stent pull wire, pusher shaft, or a combination thereof), configured to position, expand, retract, re-position, and/or remove the stent **490** from the body lumen.

[0121] FIG. 30 shows a cross-sectional view of a region of the delivery system 400 in the deployed state within the body lumen. As shown in FIG. 30, the stent 490 is expanded within the vessel by a balloon 480. To expand the deployed stent 490, the balloon 480 is coupled to inner shaft 410 and distally advanced into a lumen of the deployed stent 490 until a distal tip 415 of the inner shaft 410 is positioned near a distal end 490b of the deployed stent 490. As illustrated, the distal end 490b of the stent 490 can include radiopaque markers 490c. Additionally or alternatively, radiopaque markers 490c may be located elsewhere in the delivery system 400, or may be omitted from the delivery system 400.

[0122] FIG. 31 shows a cross-sectional view of a portion of the delivery system 400 in a deployed state with the stent 490 having protruding features 494 expanded within and piercing a portion of the vessel wall. As illustrated in FIG. 31, the balloon 480 is deployed and radially expanded to engage with and further expand the stent 490 into contact with the lumen vessel. When the stent 490 is expanded, the protruding features 494 penetrate further into the wall.

[0123] FIG. 32 shows a cross-sectional view of the delivery system 400 in a treatment state. In the treatment state, the balloon 480 has been deflated. As illustrated, the distal portion 410b of the inner shaft 410 remains in the body lumen. Following deflation of the balloon 480, the stent 490 remains expanded into contact with the lumen wall and the protruding features 494 remain penetrated into the wall. Any drugs carried by the protruding features 494 are at least partially released into the body lumen wall at the treatment state. Optionally, the stent 490 can be affixed to the balloon 480, such as by crimping the stent 490 to at least partially surround the balloon 480, such that the stent 490 is both expanded and collapsed by inflating and deflating the balloon 480, respectively.

[0124] While the stents described herein have the features shown, it will be understood that a variety of different stents and other devices can be used with the delivery systems described herein. Various features are set forth below by way of example, and not by limitation.

[0125] Regarding such stents and other devices, the material(s) for forming the frame, struts, and/or protruding features described herein can be selected based on mechanical and/or thermal properties, such as strength, ductility, hardness, elasticity, flexibility, flexural modulus, flexural strength, plasticity, stiffness, emissivity, thermal conductivity, specific heat, thermal diffusivity, thermal expansion, any of a variety of other properties, or a combination thereof. If formed from a material having thermal properties, the material can be activated to deliver thermal treatment to the desired treatment site. Regardless of the material, the frame, struts, and/or protruding features can be formed from a tube or a wire, such as a solid wire, by laser cutting or other suitable techniques. When formed from the wire, a portion of the wire can be removed by chemical etching or another suitable method to create an inner dimension of the stent.

[0126] Stents (e.g., the frame and the struts) can be sized and shaped for placement within various body lumens, including blood vessels, while not rupturing the vessel. For example, several stents and other structures can have radial strength that allows for features of the body lumen (e.g., vessel wall) to receive drugs without dissection or damage thereto. Vessels in which the stents described herein may be sized and shaped for placement include arteries, such as coronary arteries, peripheral arteries, carotid arteries, circle of willis, anterior cerebral artery, middle cerebral artery, posterior cerebral artery, any of the lenticulostriate arteries, renal arteries, femoral arteries, veins, such as cerebral veins, saphenous veins, arteriovenous fistulas, or any other vessel that may contain a treatment site. Stents can have a variety of shapes, including a cube, a rectangular prism, a cylinder, a cone, a pyramid, or variations thereof.

[0127] Stents and other structures having protruding features can include a variety of dimensions (in both the low-profile delivery state and expanded deployed state). These embodiments can provide for expansion that enables usage in a variety of situations covering a wide range of dimensions, such as to treat and/or prevent dissection. Regardless of the shape, stents can have a length of about 0.25 mm, about 0.5 mm, about 1 mm, about 2 mm, about 3 mm, about 4 mm, about

5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 10 mm, about 12 mm, about 14 mm, about 16 mm, about 18 mm, about 20 mm, about 30 mm, about 40 mm, about 50 mm, about 60 mm, about 70 mm, about 80 mm, about 90 mm, or about 100 mm. In addition, a stent shaped into a cube, a rectangular prism, or a pyramid can have a width of about 0.25 mm, about 0.5 mm, about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 10 mm, about 12 mm, about 14 mm, about 16 mm, about 18 mm, about 20 mm, about 25 mm, or about 30 mm. Moreover, a stent shaped into a cylinder or a cone can have a diameter of about 0.25 mm, about 0.5 mm, about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 10 mm, about 12 mm, about 14 mm, about 16 mm, about 18 mm, about 20 mm, about 25 mm, about 30 mm, about 35 mm, about 40 mm, or about 50 mm. The width or the diameter of the stent can decrementally decrease along a length of the stent. In addition, the stent can be sized and shaped to prepare the body lumen for certain procedures, such as a stent placement procedure.

[0128] A stent and/or other expandable structures in the expanded state can have a cross-sectional dimension of about 2 mm to about 10 mm, inclusive of the expanded protruding features. For example, a frame can have a cross-sectional dimension of about 1 mm to about 9 mm and the protruding features can each have a length from about 0.1 mm to about 1.5 mm. In some embodiments, the stent has an overall cross-sectional dimension of about 4 mm with the frame having a cross-sectional dimension of about 2 mm and the protruding features each having a length of about 1 mm. In some embodiments, the stent has an overall cross-sectional dimension of about 6 mm with the frame having a cross-sectional dimension of about 4 mm and the protruding features each having a length of about 1 mm. In further embodiments, the protruding features can have a plurality of lengths such that the length of the protruding features of a stent or other expandable structure differs. For example, a stent can include protruding features having a length of about 0.2 mm, about 0.5 mm, and about 1 mm.

[0129] Profiles of the stents or other structures can be sized such that the stents or other structures are compatible with a wide range of catheter sizes. Embodiments in accordance with the present technology can include stents or other structures designed to receive a guidewire, such as guidewires having a diameter of 0.010, 0.014, 0.018, 0.035, or 0.038 inch. In several embodiments, the stent or scaffold structure can be sized and designed for delivery via a micro-catheter that it is pushed through. In some embodiments, stents or structures can be incorporated into a delivery system, including modular or single unit delivery systems.

[0130] Stents and other structures described herein can include a marking for visualization of the stent within the body lumen, such as one or more radiopaque markers. The radiopaque markers can be formed from Clearfil Photo Core PLT®, tantalum, titanium, tungsten, barium sulfate, and zirconium oxide, or another suitable radiopaque marking. The markings can be formed on a proximal portion of the stent, a distal portion, an intermediate portion, or a combination thereof. The markings can be a band, a coil, a clip, filled into one or more portions of a tube in the stent, plated onto one or more portions of the stent, or a combination thereof. Regardless of the type of marking, the marking can be coined, swaged, wrapped, or encased along, or onto any portion of the stent.

[0131] Stents and other structures can be flexible enough to track through various anatomical features, including those having a curvature. The flexible properties of the stent and other structures can be provided by the material from they are formed. In addition, flexible properties can also be provided by fracturing one or more of the members engaging with and extending between two or more rows of struts. Additionally, the stent or other structure can be readily deployed and expanded, and retracted and contracted. The stent or other structure can also be readily repositioned within a vessel or other body lumen.

[0132] In several embodiments, a drug-eluting compound is coated onto at least a portion of the protruding features, the frame, the struts, and/or the balloon. The coating can be any suitable

coating known to one of ordinary skill in the art suitable to deliver the drug to the wall. For example, suitable coatings include, but are not limited to a snow coating or a crystalline coating having edges configured to remain in the wall. The drug-eluting compound can be a synthetic or biological polymer coated into a variety of different patterns and thicknesses suitable for delivering the drug contained therein. In other embodiments, the protruding features themselves may be composed of drug-eluting materials. The drug carried by the drug-eluting compound and/or the protruding features in accordance with the present technology can be any drug suitable for treating the treatment site in which the stent will be placed and may or may not include an excipient. For example, the drug can be an anti-proliferative, an anti-neoplastic, a migration inhibitor, an enhanced healing factor, an immunosuppressive, an anti-thrombotic, a blood thinner, or a radioactive compound. Examples of anti-neoplastics include, but are not limited to, sirolimus, tacrolimus, everolimus, leflunomide, M-prednisolone, dexamethasone, cyclosporine, mycophenolic acid, mizoribine, interferon, and tranilast. Examples of anti-proliferatives include, but are not limited to, taxol/paclitaxel, actinomycin, methotrexate, angiopeptin, vincristine, mitomycin, statins, c-myc antisense, Abbot ABT-578, RestinASE, 2-chloro-deoxyadenosine, and PCNA ribozyme. Examples of migration inhibitors, but are not limited to, include batimistat, prolyl hydroxylase, halofunginone, c-preteinase inhibitors, and probucol. Examples of enhanced healing factors include, but are not limited to, BCP 671, VEGF, estradiols, NO donor compounds, and EPC antibodies. Examples, of radioactive compounds include, but are not limited to, strontium-89 chloride (Metastron®), samarium-153 (Quadramet®), radium-223 dichloride (Xofigo®), yttrium-90, and iodine-131. In some embodiments, the drug-eluting compound and/or the protruding features can carry more than one drug.

[0133] In some embodiments, the protruding features can include textured (e.g., ribbed) surfaces which is expected to provide greater surface area for drug-delivery. Moreover, any protruding features can include a textured surface such as a ribbed surface (vertical, horizontal, radial, or circular relative to a longitudinal plane of the protruding feature), a cross-hatched surface, an isotropic surface, or other surface types suitable for providing greater surface area for drug-delivery.

[0134] The protruding features can be sized and shaped to engage with and/or penetrate an occlusion, a neointima, an intima, an internal elastic lamina (IEL) a media, an external elastic lamina (EEL), an adventitia, or a combination thereof. The protruding features can also be sized and shaped to engage with and/or penetrate a tissue and/or structure adjacent to the body lumen in which the stent is to be placed while not rupturing the body lumen. For example, the stent can include square protruding features sized and configured to penetrate into the intima and/or the media of a body lumen, pointed protruding features sized and configured to penetrate and extend into the media, and/or the IEL. In addition, protruding features can be configured to bend in one or more directions relative to a longitudinal axis of the stent to engage with and/or penetrate a portion of the body lumen described herein. In several embodiments, the protruding features can penetrate deeper into the wall of a diseased body lumen, such as a vessel, compared to a stent lacking protruding features. In addition, the stent can allow for blood to flow even while in the expanded position and with drug-eluting on-going.

[0135] Various protruding features described herein can deliver drugs deeper into a vessel wall than possible via angioplasty balloons or other existing devices. In addition to carrying one or more drugs for treatment of the site, the protruding features can also carry a molecule suitable for degrading a portion of the occlusion, neointima, and/or intima to allow the protruding features to penetrate deeper in to the vessel wall than without the molecule. For example, the molecule suitable for degradation can be an enzyme, such as elastase, collagenase, or a proteinase, such as, metalloproteinases, serine proteinases, cysteine proteinases, extracellular sulfatases, hyaluronidases, lysyl oxidases, lysyl hydroxylases, or a combination thereof.

[0136] Further, it will also be appreciated that stents can carry one or more protruding features on

one or more portions of the stent. For example, the stents can carry about 5 protruding features, about 10 protruding features, about 15 protruding features, about 20 protruding features, about 30 protruding features, about 40 protruding features, about 50 protruding features, about 60 protruding features, about 70 protruding features, about 80 protruding features, about 90 protruding features, or about 100 protruding features. The protruding features can be carried by the frame, the struts, or a combination thereof. The number of protruding features can vary depending upon, for example, the target treatment site, the type of drug being delivered, and size of the stent, etc. In addition, the protruding features carried by the stent can be different types of the protruding features disclosed herein.

[0137] In some embodiments, once positioned against a body lumen wall (e.g., a vessel wall), tissue and/or fluid can interact with the protruding feature to dissolve the drug and selectively release it from the reservoir. In other embodiments, the protruding feature can be configured to deliver the drug via a variety of means once the stent is expanded. Protruding features are accordingly expected to provide an effective means for selectively delivering a drug to a desired location, while reducing inadvertent loss or release of drugs. In other embodiments, the stent can include more than one protruding feature, or a protruding feature having more than one reservoir. In several embodiments, the stent including protruding features can have the protruding feature, such as the coating or the reservoir, concealed (e.g., recessed) until the stent is positioned at the treatment site. Once positioned at the target site, the protruding feature can be revealed (e.g., expanded/projected, etc.) during and/or after expansion of the stent. This is expected to reduce any loss of the drug carried by the protruding feature during delivery to the treatment site.

[0138] In some embodiments, the stents can further include a material (e.g., PTFE, Dacron, polyamides, such as nylon and/or polyurethane based materials, silicone, etc.) positioned over a stent, scaffold or other structure having protruding features covering at least a portion of the outer surface area. In some embodiments, the material covers the entire outer surface area. The material can be a mesh or a braid. In some embodiments, the material can be configured to increase a surface area of the stent useful for providing additional surface area of the stent for coating with a drug. In other embodiments, the material can further be configured to allow blood flow through the inner diameter of the stent and/or limit blood flow to an outer dimension of the stent. In additional embodiments, the material can create a barrier between fluid flow (e.g., blood flow) and the drug-delivery locations. In addition, the material can be configured to prevent debris from the wall of the body lumen from entering the bloodstream. In such embodiments, the associated systems and devices can be used for temporary dissection tacking or coverage of a region that may have been perforated during a procedure.

[0139] The embodiments described herein provide delivery systems for one or more structures having a means for delivering drugs to a specific region within a body lumen, such as the vasculature, while still allowing fluid (e.g., blood) to flow through the treatment area where the structure has been placed and/or other devices or treatment means within the adjacent body lumen. In some embodiments, the fluid is temporarily prevented from flowing through the treatment area while one or more regions of systems is delivered, deployed, positioned, and/or removed from the body lumen. In addition, the delivery systems can be configured to prepare the body lumen for treatment, by raking the stent, pulling the stent, turning the stent, or a combination thereof, proximal or distal to the treatment site. In other embodiments, the delivery systems can be configured to rotate the stent when mechanical force is applied.

[0140] The systems disclosed herein can provide for adjustment, recapture, and/or redeployment of the associated stents or other structures, and/or deployment of a different stent or other structure, allowing a practitioner to more effectively to treat a desired region more accurately and deliberately. In several embodiments, the stent or other delivery structure can be deployed for a temporary period (e.g., for less than 24 hours), and then retracted and removed. In these embodiments, the protruding features can engage with and/or pierce the lumen wall and remain

therein after the stent or other delivery structure is removed, or can be retracted and removed with the stent or other delivery structure. The stent can be configured to self-expand, or partially self-expand, when deployed from the delivery system and also be configured to further expand within the body lumen when the balloon is expanded therein. The stent can also be configured to post-dilate when removed from the body lumen. In other embodiments, the stent or other delivery structure can be deployed for a long-term temporary period (e.g., for less than 2 weeks, less than one month, less than 6 months, less than one year), and then retracted and removed. In some embodiments, a different stent or delivery structure can be deployed after a first stent or delivery structure has been retracted and removed. The duration of deployment and duration after removal before deployment of the different stent or delivery structure can vary from minutes, to hours, to days, to weeks, to months, or to years. In these embodiments, removal of the first stent or delivery structure and deployment of a different stent or delivery structure can occur once, twice, three times, four times, five times, six times, seven times, eight times, nine times, or ten times. Moreover, the embodiments described herein can allow for a lower profile system than currently available systems.

[0141] In the embodiments described herein and other embodiments configured in accordance with the present technology, stents and other expandable structures may include non-protruding features, such as deployable and/or expandable features, that are not configured for delivering a drug to a target location. For example, stents and other expandable structures configured in accordance with the present technology can include one or more protruding features, one or more non-protruding features, or combinations thereof.

[0142] While many embodiments of the stents and/or structures described herein include stents, additional embodiments of the expandable elements, such as stents and/or structures, can include non drug-eluting stents and/or non drug-eluting structures. In these embodiments, the non drug-eluting stents may include one or more protruding members, such as spikes. The spikes can be configured to engage with and/or penetrate a portion of the body lumen or vessel. For example, the spikes can penetrate the vessel wall, thereby reducing and/or eliminating an elasticity of the vessel wall. In these embodiments, the protruding members can be configured to prevent the vessel wall from progressing inward toward the body lumen and restricting and/or constricting flow therein. The protruding members can be integrally formed with the struts, or disposed on the surface of the struts, extending radially outward from the struts toward the target tissue.

[0143] Various examples of aspects of the disclosure are described below as clauses for convenience. These are provided as examples, and do not limit the subject technology.

[0144] Clause A: a delivery system comprising: an outer shaft; an inner shaft slideably disposed within the outer shaft and comprising an inflatable balloon; a guidewire slideably disposed within the inner shaft; and a stent disposed around the balloon and fixedly coupled to the inner shaft.

[0145] Clause B: a method for delivering a stent within a body lumen of a patient, the method comprising: delivering an outer shaft ensheathing an inner shaft and a stent to a target treatment site within the body lumen of the patient, the stent being disposed around a balloon of the inner shaft and fixedly coupled to the inner shaft; proximally retracting the outer shaft to at least partially unsheath the stent; radially expanding the stent to an expanded state by expanding the balloon; and piercing through a portion of a wall of the body lumen with one or more protruding features of the stent.

[0146] Clause C: a delivery system comprising: an outer shaft; an inner shaft slideably disposed within the outer shaft and comprising an inflatable balloon on a distal portion of the inner shaft; a guidewire slideably disposed within the inner shaft; and a stent slideably disposed within the outer shaft and on a proximal portion of the inner shaft that is proximal to the balloon, the stent being connected to a stabilizing wire that is slideably disposed within the outer shaft.

[0147] Clause D: a method for delivering a stent within a body lumen of a patient, the method comprising: delivering an outer shaft ensheathing an inner shaft and a stent to a target treatment site

within the body lumen of the patient, the stent being slideably disposed within the outer shaft and on a proximal portion of the inner shaft that is proximal to a balloon of the inner shaft; proximally retracting the outer shaft to at least partially unsheath the stent and the balloon; moving the inner shaft proximally relative to the stent until the stent is axially aligned with the balloon; radially expanding the stent to an expanded state by expanding the balloon; and piercing through a portion of a wall of the body lumen with one or more protruding features of the stent.

[0148] Clause E: a delivery system comprising: an outer shaft; an inner shaft slideably disposed within the outer shaft and comprising an inflatable balloon on a proximal portion of the inner shaft; a guidewire slideably disposed within the inner shaft; and a stent slideably disposed within the outer shaft and on a distal portion of the inner shaft that is distal to the balloon, the stent being connected to a stabilizing wire that is slideably disposed within the outer shaft.

[0149] Clause F: a method for delivering a stent within a body lumen of a patient, the method comprising: delivering an outer shaft ensheathing an inner shaft and a stent to a target treatment site within the body lumen of the patient, the stent being slideably disposed within the outer shaft and on a distal portion of the inner shaft that is distal to a balloon of the inner shaft; proximally retracting the outer shaft to at least partially unsheath the stent and the balloon; moving the inner shaft distally relative to the stent until the stent is axially aligned with the balloon; radially expanding the stent to an expanded state by expanding the balloon; and piercing through a portion of a wall of the body lumen with one or more protruding features of the stent.

[0150] Clause G: a method for delivering a stent within a body lumen of a patient, the method comprising: delivering an outer shaft ensheathing an inner shaft and a stent to a target treatment site within the body lumen of the patient, the stent being disposed around a balloon of the inner shaft and fixedly coupled to the inner shaft; proximally retracting the outer shaft to partially unsheath the stent; radially expanding a first length of the stent to an expanded state by expanding the balloon while a portion of the stent is within the outer shaft until one or more protruding features of the stent pierces through a first portion of a wall of the body lumen; proximally retracting the outer shaft to further unsheath the stent; and radially expanding a second length of the stent to an expanded state by expanding the balloon until one or more protruding features of the stent pierces through a second portion of the wall of the body lumen.

[0151] One or more of the above clauses can include one or more of the features described below. It is noted that any of the following clauses may be combined in any combination with each other, and placed into a respective independent clause, e.g., clause A, B, C, D, E, F, or G.

[0152] Clause 1: a connector at a proximal end of the inner shaft, the guidewire extending through the connector, the connector comprising a port in fluid communication with the balloon; and an outer shaft hub at a proximal end of the outer shaft, the inner shaft extending through the outer shaft hub.

[0153] Clause 2: incrementally spaced markers on a proximal portion of the inner shaft, wherein the outer shaft hub is slidable over the inner shaft along the proximal portion.

[0154] Clause 3: a locking member configured to lock the outer shaft hub to a proximal portion of the inner shaft such that a position of the outer shaft relative to the inner shaft is maintained when the balloon is inflated.

[0155] Clause 4: the balloon is inflatable through a lumen of the inner shaft that contains the guidewire.

[0156] Clause 5: a stiffening wire positioned radially between the inner shaft and the outer shaft.

[0157] Clause 6: the stent comprises: a radially expandable cylindrical frame comprising struts; and protruding features carried by one or more struts.

[0158] Clause 7: the stent is fixedly coupled to the inner shaft by an anchor portion that extends about at least a portion of the inner shaft.

[0159] Clause 8: the anchor portion is coupled to the inner shaft on a proximal side of the balloon.

[0160] Clause 9: the balloon comprises multiple segments at different axial positions along a length

of the inner shaft, the multiple segments each being independently inflatable.

[0161] Clause 10: deflating the balloon; advancing the outer shaft over the stent; and removing the stent from the body lumen.

[0162] Clause 11: a connector at a proximal end of the inner shaft, the guidewire extending through the connector, the connector comprising a port in fluid communication with the balloon; and an outer shaft hub at a proximal end of the outer shaft, the inner shaft and the stabilizing wire extending through the outer shaft hub.

[0163] Clause 12: the target treatment site is a first target treatment site, the method further comprising before proximally retracting the outer shaft to further unsheath the stent, repositioning the stent to a second target treatment site.

[0164] Clause 13: the second length of the stent includes the first length of the stent.

[0165] Clause 14: before proximally retracting the outer shaft to further unsheath the stent, deflating the balloon and resheathing the stent.

[0166] Clause 15: before radially expanding the first length of the stent, locking the outer shaft relative to the inner shaft.

[0167] Clause 16: before radially expanding the second length of the stent, locking the outer shaft relative to the inner shaft.

[0168] A reference to an element in the singular is not intended to mean one and only one unless specifically so stated, but rather one or more. For example, “a” module may refer to one or more modules. An element preceded by “a,” “an,” “the,” or “said” does not, without further constraints, preclude the existence of additional same elements.

[0169] Headings and subheadings, if any, are used for convenience only and do not limit the invention. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0170] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0171] A phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, each of the phrases “at least one of A, B, and C” or “at least one of A, B, or C” refers to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0172] It is understood that the specific order or hierarchy of steps, operations, or processes disclosed is an illustration of exemplary approaches. Unless explicitly stated otherwise, it is understood that the specific order or hierarchy of steps, operations, or processes may be performed in different order. Some of the steps, operations, or processes may be performed simultaneously.

The accompanying method claims, if any, present elements of the various steps, operations or processes in a sample order, and are not meant to be limited to the specific order or hierarchy presented. These may be performed in serial, linearly, in parallel or in different order. It should be understood that the described instructions, operations, and systems can generally be integrated together in a single software/hardware product or packaged into multiple software/hardware products.

[0173] In one aspect, a term coupled or the like may refer to being directly coupled. In another aspect, a term coupled or the like may refer to being indirectly coupled.

[0174] Terms such as top, bottom, front, rear, side, horizontal, vertical, and the like refer to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, such a term may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.

[0175] The disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology. The disclosure provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles described herein may be applied to other aspects.

[0176] All structural and functional equivalents to the elements of the various aspects described throughout the disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.

[0177] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.

[0178] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language of the claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

Claims

1. A system comprising: an outer shaft; an inner shaft slidably disposed within the outer shaft and including an inflatable balloon; a guidewire slidably disposed within the inner shaft; and a stent disposed around and axially aligned with a portion of the balloon, the stent including a cylindrical frame and protruding features extending from the cylindrical frame, the stent being fixedly coupled to the inner shaft by a stabilizing wire connecting a proximal end of the stent to a portion of the

inner shaft that is proximal to the balloon, the stent being retractable by sliding the inner shaft and the stabilizing wire within the outer shaft until the outer shaft is over the stent.

2. The system of claim 1, further comprising: a connector at a proximal end of the inner shaft, the guidewire extending through the connector, the connector including a port in fluid communication with the balloon; and an outer shaft hub at a proximal end of the outer shaft, the inner shaft extending through the outer shaft hub.

3. The system of claim 2, further comprising incrementally spaced markers on a proximal portion of the inner shaft, wherein the outer shaft hub is slidable over the inner shaft along the proximal portion.

4. The system of claim 2, further comprising a locking member configured to lock the outer shaft hub to a proximal portion of the inner shaft such that a position of the outer shaft relative to the inner shaft is maintained when the balloon is inflated.

5. The system of claim 1, wherein the balloon is inflatable through a lumen of the inner shaft that contains the guidewire.

6. The system of claim 1, further comprising a stiffening wire positioned radially between the inner shaft and the outer shaft.

7. The system of claim 1, wherein the balloon includes multiple segments at different axial positions along a length of the inner shaft, the multiple segments each being independently inflatable.

8. The system of claim 1, wherein the stent further includes an intermediate portion that extends from the cylindrical frame and about at least a portion of the inner shaft, the stabilizing wire extending from the intermediate portion and including an anchor portion coupled to the inner shaft.

9. The system of claim 8, wherein the intermediate portion includes multiple struts each extending from the cylindrical frame to the anchor portion.

10. The system of claim 1, wherein an anchor portion of the stabilizing wire extends about at least a portion of the inner shaft.

11. A method comprising: delivering over a guidewire an outer shaft ensheathing an inner shaft and a stent to a target treatment site within a body lumen of a patient, the guidewire being slidably disposed within the inner shaft, the stent being disposed around and axially aligned with a portion of a balloon of the inner shaft, the stent including a cylindrical frame and protruding features extending from the cylindrical frame, the stent being fixedly coupled to the inner shaft by a stabilizing wire connecting a proximal end of the stent to a portion of the inner shaft that is proximal to the balloon; proximally retracting the outer shaft to at least partially unsheath the stent and the balloon; radially expanding the stent to an expanded state by expanding the balloon; and piercing through a portion of a wall of the body lumen with one or more protruding features of the stent.

12. The method of claim 11, wherein during the delivering: the guidewire is extending through a connector at a proximal end of the inner shaft, the connector including a port in fluid communication with the balloon; and the inner shaft is extending through an outer shaft hub at a proximal end of the outer shaft.

13. The method of claim 11, wherein the balloon is expanded by inflating the balloon through a lumen of the inner shaft that contains the guidewire.

14. The method of claim 11, further comprising: collapsing the balloon; and distally advancing the outer shaft to sheath the stent and the balloon.

15. The method of claim 11, wherein, while distally advancing the outer shaft to sheath the stent and the balloon, an axial position of the stent is maintained by the inner shaft and the stabilizing wire.

16. The method of claim 11, wherein an anchor portion of the stabilizing wire extends about at least a portion of the inner shaft.

17. A system comprising: an outer shaft; an inner shaft slidably disposed within the outer shaft and

including an inflatable balloon on the inner shaft; and a stent being slidably disposed within the outer shaft and extending about a first portion of the inner shaft, the stent being fixedly coupled to the inner shaft by an anchor portion that is connected to a proximal end of the stent and extends about at least a second portion of the inner shaft that is proximal to the inflatable balloon.

18. The system of claim 17, wherein the stent includes: a radially expandable cylindrical frame including struts; and protruding features carried by one or more struts.

19. The system of claim 17, further comprising a stiffening wire positioned radially between the inner shaft and the outer shaft.

20. The system of claim 17, further comprising a guidewire slidably disposed within a lumen of the inner shaft.
